Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			award (1) for either of following energy of the *d*-electrons are similar successive ionisation energies are similar	1			1		
	(b)	(i)		blue (1) the frequency of light that is <u>not absorbed</u> is in this region (1)		2		2		
		(ii)	I	a (small) molecule/ion with a lone pair (that can bond to a transition metal ion)	1			1		
			II	award (1) for any of following lone pair from ligand combines with empty orbital lone pair from ligand can combine with electron deficient transition metal atom sharing of a pair of electrons which are both from ligand forms a coordinate bond (1)	2			2		
	(c)	(i)		$2I^- \rightleftharpoons I_2 + 2e^-$		1		1		
		(ii)		$VO_2^+ + 2H^+ + e^- \rightleftharpoons VO^{2+} + H_2O$		1		1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		catalyst in a different physical state from the reactants / reaction	1			1		
		(ii)	I	$[2 \times \Delta_f H\,(SO_3)] - [2 \times \Delta_f H\,(SO_2)] = -197$ (1) $\Delta_f H\,(SO_3) = -395.5$ kJ mol^{-1} (1)		2		2	2	
			II	$T = \frac{\Delta H}{\Delta S}$ (1) $T = 1053$ K (1) student is incorrect must give reason ΔG is negative below this temperature so the reaction is feasible at a lower temperature (1)		2	1	3	2	
			III	initial concentrations of SO_2 and $O_2 = 0.020$ mol dm^{-3} (1) equilibrium concentrations – both needed for (1) $[SO_2] = 0.006$ mol dm^{-3} $[O_2] = 0.013$ mol dm^{-3} $K_c = \frac{(0.014)^2}{(0.006)^2 \times (0.013)} = 419$ (1) mol^{-1} dm^3 (1)		4		4	4	
				Question 8 total	**5**	**12**	**1**	**18**	**8**	**0**